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# The Removal of Model Viruses, Poliovirus type 1 and Canine Parvovirus, During the Purification of Human Albumin using Ion-exchange Chromatographic Procedures

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**Abstract.** The manufacturing process for albumin in Australia is based primarily on ion-exchange chromatography. The capacity of ion-exchange matrices to remove non-enveloped viruses (canine parvovirus and poliovirus type 1) was assessed using a scaled-down chromatographic process which was shown to yield product meeting purity criteria set for the manufacturing process. Poliovirus type 1 and canine parvovirus were added at one tenth the volume of desalted and delipidated Supernatant II + III produced by traditional Cohn Fractionation from human plasma before the material was applied to DEAE and CM ion-exchangers connected in series. Samples were taken at equilibration, wash, elution and regeneration steps and the log clearance and reduction of the viruses calculated. The mean clearance and reduction factors for viral load of poliovirus type 1 were 5·3 logs and 3·2 logs, respectively and 1·8 logs and 1·8 logs for canine parvovirus.



## Introduction

Chromatographic procedures are now used at some stage during the downstream processing or purification of the majority of recently developed biopharmaceuticals, such as monoclonal immunoglobulins produced by hybridoma technology and recombinant proteins derived from microbes or cell culture systems. The major components of human plasma, immunoglobulin and albumin, however are still purified principally by cold ethanol fractionation procedures based on methods originally developed by Cohn and coworkers in 1946.[1] A chromatography procedure for the purification of albumin was first developed almost 20 years[2] ago but has not gained widespread approval among plasma manufacturers because of the cost associated with changing existing manufacturing facilities and registering a new albumin product. There are, however, significant advantages in chromatographic methods over the traditional Cohn method including: ease of automation of chromatographic processes; production of higher purity product; and the potential to isolate other plasma proteins which may otherwise be denatured. In order to change manufacturing processes it is important that the fate of potential blood-borne virus contaminants in the process be understood. This paper seeks to assess virus clearance achieved during the chromatographic purification of albumin,[3] using two model viruses, poliovirus type 1 (PV1) and canine parvovirus (CPV). These non-enveloped viruses were used as surrogate markers for closely related hepatitis A (HAV) and human parvovirus B19 (B19), with B19 not able to be readily assayed in the laboratory. Outbreaks of infections with HAV and B19 have been attributed to the use of contaminated blood products prepared under adequate conditions to inactivate enveloped virus such as hepatitis B virus (HBV), hepatitis C virus (HCV) and human immunodeficiency virus (HIV).[4,5] In this study a scaled-down version of the ion-exchange production step was challenged with virus. The scaled-down system was shown to generate product within the purity limits set for the large-scale process.

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